AYLA NAJAM KHAN

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PROFESSIONAL SUMMARY

Computer Programming & Analysis student with a **4.17 GPA** and experience building 4+ software and data applications across Python, Java, C++, SQL, and web technologies, including an AI/NLP resume matching tool. Managed business data in SQL for 2+ years in a remote role. Skilled in REST APIs, data validation, debugging, functional testing, and collaborative problem solving.

TECHNICAL SKILLS

Languages: Python, Java, C++, Go, JavaScript, SQL, HTML5, CSS3
AI / Data: Scikit-learn, NLP, TF-IDF, Cosine Similarity, Machine Learning, Pandas, NumPy, Jupyter Notebook
Databases & APIs: Microsoft SQL Server, Microsoft Access, Relational Databases, JSON, REST APIs, API Documentation
Development & QA: Git, GitHub, Visual Studio, VS Code, Debugging, Input Validation, Functional Testing, OOP, Claude Code, Codex
Systems & Networking: Windows, Linux/Ubuntu, TCP/IP, DNS, DHCP, Cisco Packet Tracer, Microsoft Visio

EDUCATION

Advanced Diploma — Computer Programming & Analysis — London, Ontario
Fanshawe College — Expected 2028

- **GPA:** Semester 1: 3.96 | Semester 2: 4.17
- **Coursework:** Object-Oriented Programming, Data Structures & Algorithms, Advanced Databases, Networking, Client/Server Computing, Operating Systems, Web Development

TECHNICAL PROJECTS

MatchWise AI — Intelligent Resume Analyzer
Python • Flask • NLP • Scikit-learn • TF-IDF • Cosine Similarity • HTML/CSS

- Built an AI/NLP resume-matching platform supporting **3 file formats** (PDF, DOCX, TXT) with drag-and-drop uploads and 10 MB file validation.
- Engineered a matching system covering **40+ technical skill categories** across 2 scoring dimensions: skill alignment and NLP-based content similarity.
- Implemented **3 specialized AI agents** for job-requirement analysis, resume evaluation, and targeted resume-improvement recommendations.
- Applied TF-IDF and cosine similarity using a **70/30 weighted model** to generate a 0–100% match score, identify matched and missing skills, and calculate the remaining score gap.
- Used Claude Code to assist with debugging, code refinement, and feature development.

London International Airport Database
SQL • Microsoft Access

- Designed a relational database of **8 linked tables** covering flights, passengers, and staff, enforcing referential integrity across **12 relationships**.
- Wrote **15+ SQL queries** using multi-table joins and filtering to answer operational questions including gate utilization and passenger manifests.
- Built **6 data-entry forms and 6 reports** with field-level validation, reducing invalid entries during system testing and making recurring information easier to retrieve.

Newcomb-Benford Statistical Analysis Program
C++

- Built a C++ statistical-analysis tool capable of parsing datasets containing up to **10,000 numeric records** and calculating leading-digit distributions against Benford's Law.
- Implemented input validation and file-handling routines that reject malformed rows without crashing, verified across **8+ test datasets**.
- Validated statistical and Newcomb-Benford outputs against expected results, achieving a **100% match**.

Multiplayer Craps Game
Java • Swing • OOP

- Developed an event-driven Java Swing game supporting **2–6 players** with automated turn rotation, betting, point/pass rules, and winner detection.
- Implemented player-state management with a **$100 starting balance** per player and organized functionality into **6 reusable classes**.
- Tested **20+ gameplay scenarios** covering betting, balance updates, turn changes, point/pass rules, and win conditions, resolving logic and interface issues before submission.

EXPERIENCE

Administrative & Digital Support Assistant — Pakistan | Remote
Universal Fumigation Co. — Sep 2023 – Oct 2025

- Maintained and validated SQL records for **500+ customers** over 2+ years, identifying duplicate, incomplete, and inconsistent data before it entered business workflows.
- Diagnosed and resolved **10+ data and workflow issues per month**, cutting duplicate and incomplete record entries by **20%**.
- Managed correspondence and deadlines independently for a **remote team of 5**, ensuring timely communication and task completion.

Social Media Manager — Pakistan | Remote
Sowear — Apr 2023 – Aug 2025

- Analyzed engagement data across **3 social platforms**, refining content strategy to drive **30% follower growth** and increase engagement by **25%**.
- Managed a publishing calendar of **5+ posts per week**, maintaining **100% on-time publishing** with zero missed scheduled posts.

CERTIFICATIONS

- **Python for Data Science, AI & Development** | IBM, Coursera
- **The Bits and Bytes of Computer Networking** | Google, Coursera